AUTOMATED DETECTION, TRACKING AND TACTICAL ANALYSIS IN BROADCAST FOOTBALL VIDEO

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List of Abbreviations

HOTA	Higher Order Tracking Accuracy
IDF1	Identification F1
IoU	Intersection over Union
MAE	Mean Absolute Error
mAP	mean Average Precision
MOTA	Multi-Object Tracking Accuracy

Abstract

TODO: my abstract (to be done at the end of the report)

Declaration

No portion of the work referred to in this report has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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Chapter 1

Introduction

1.1 Motivation and Context

The use of data and analytics in professional football has expanded substantially in recent years, with clubs and national federations increasingly incorporating analytical workflows into performance analysis and decision-making [15]. In particular, positional and tracking data are valuable because they enable richer descriptions of player movement, team organisation and tactical behaviour than are possible from traditional notational analysis alone [6].

Despite this growing importance, high-quality football data remain difficult to obtain. Advanced event and tracking data are often proprietary, expensive, or dependent on substantial manual collection, which limits their accessibility outside elite settings [23]. This creates a strong motivation to investigate alternative ways of generating structured football data from more widely available inputs.

Broadcast video is a particularly attractive source for this purpose. Unlike specialised tracking systems or instrumented capture environments, broadcast footage is widely available and already captures much of the spatial and contextual information required for match analysis. However, extracting reliable structured data from broadcast footage is technically challenging. Prior work on broadcast soccer analysis has shown that the problem involves several tightly coupled sub-tasks, including player detection, tracking, team assignment and field registration, all of which are affected by factors such as camera motion, occlusion, scale variation and motion blur [14, 22]. Earlier work has also shown that tracking football players directly from TV broadcasts is feasible, but only by addressing a difficult combination of camera estimation, player localisation and identity disambiguation under realistic match conditions [4].

This project is motivated by the question of how far a modern, modular computer vision pipeline can go in turning raw broadcast football video into useful tactical and spatial information. Rather than focusing on a single isolated task, the project investigates an end-to-end pipeline that combines detection, multi-object tracking, unsupervised team assignment, pitch registration, homography-based projection and event reasoning. The intended outcome is a system that can transform broadcast footage into structured analytical outputs such as annotated tracking video, bird's-eye tactical views, heatmaps, match statistics and offside analysis. The overall aim is not to replace professional-grade tracking systems, but to investigate how far a modular vision pipeline can go in extracting practically useful information from widely available video data.

1.2 Aims and Objectives

The overall aim of this project is to design and implement an automated computer vision pipeline that analyses broadcast football video and produces structured tactical and spatial information without requiring specialised capture systems or manual team labels.

To achieve this aim, the project pursues the following objectives:

- develop and train an object detection model capable of identifying the main on-pitch entities, namely players, goalkeepers, referees, and the ball;
- implement a multi-object tracking stage that maintains consistent identities for detected entities across video frames;
- design an unsupervised team assignment method that groups players by team using visual appearance cues rather than manual class labels;
- detect pitch keypoints and estimate a homography that maps image-space positions into a two-dimensional pitch representation;
- build a dedicated ball detection component with temporal smoothing to improve robustness against the ball's small size, rapid motion, and frequent occlusion;
- generate higher-level tactical outputs, including bird's-eye visualisations of player positions and per-player heatmaps of spatial activity;

- implement offside analysis by combining ball possession changes, projected player positions, and the offside rule in pitch-space coordinates;
- compute derived match statistics, including player speed, distance covered, pass counts, and team possession; and
- integrate these components into a modular, configurable end-to-end pipeline that processes broadcast footage and produces multiple analytical outputs.

Together, these objectives define a project that spans several areas of computer vision and machine learning, and whose success depends not only on individual component quality but also on the reliability of their integration into a single coherent system.

1.3 Evaluation Strategy

The project will be evaluated at both *component level* and *system level*, since this is an end-to-end broadcast football analysis pipeline whose final performance depends not only on the accuracy of individual modules, but also on how reliably they interact [22]. In general, good performance means that individual components achieve strong task-appropriate metrics, while the integrated pipeline produces coherent, tactically meaningful outputs from broadcast footage.

1.3.1 Component-Level Evaluation

Object detection. The main object detector will be evaluated using standard detection metrics, with mean Average Precision (mAP) at Intersection over Union (IoU) thresholds from 0.50 to 0.95 (*mAP50–95*) as the primary measure and *mAP50*, precision, and recall as supporting metrics. Per-class results for players, goalkeepers, referees, and the ball will be reported because these classes differ significantly in difficulty. A confusion matrix will also be included to highlight the main sources of classification error.

Dedicated ball detection. Because the ball is the smallest and most challenging object in broadcast footage, the dedicated ball detector will be evaluated separately. Held-out validation results obtained during training will be reported using the same detection metrics. In addition, performance will be assessed on a manually labelled short clip to measure detection rate and false positive rate under realistic broadcast conditions. This will distinguish offline model performance from practical in-video performance.

Pitch keypoint detection and homography. Pitch keypoint detection will be evaluated using pose-specific metrics, again with *mAP50–95* as the main measure. To assess the practical quality of the estimated homography, a small reprojection-error experiment will be conducted on selected frames. Manually identified pitch landmarks will be transformed into pitch coordinates and compared with their expected locations, providing a geometric measure of field registration accuracy, which is a standard concern in sports-field calibration and registration tasks [21].

Multi-object tracking. Tracking performance will be evaluated on a manually annotated short segment using standard multi-object tracking metrics such as Higher Order Tracking Accuracy (*HOTA*), Multi-Object Tracking Accuracy (*MOTA*), and Identification F1 (*IDF1*). These are appropriate because they capture both detection quality and temporal identity consistency [16, 5]. The number of identity switches will also be reported to provide an interpretable measure of tracking stability. Since dense annotation of full matches is outside the scope of this project, formal tracking evaluation will be limited to a short labelled subset.

Team assignment. The unsupervised team assignment stage will be evaluated on a manually labelled sample of frames using precision, recall, and *F1 score*, which are more informative than accuracy alone for this binary task. A small ablation study will compare results before and after temporal smoothing to test whether majority-vote smoothing improves label stability. Goalkeeper-to-team resolution, which assigns each detected goalkeeper to the nearest team cluster, will be assessed as part of this evaluation by checking whether goalkeepers are consistently assigned to the correct team.

Offside analysis. Offside detection will be evaluated at event level using a manually labelled set of pass events containing both offside and non-offside situations. Predictions will be compared against these labels using event-level precision, recall, and *F1 score*. In addition, all detected offside events in the final outputs will be manually checked frame by frame to identify both successful detections and common failure cases.

Team possession. Team possession will be evaluated on a short manually labelled segment by comparing predicted possession percentages with hand-annotated ground truth using Mean Absolute Error (*MAE*). This provides a direct measure of how closely the estimated possession values match the manually observed distribution of possession over time.

Derived statistics and higher-level outputs. Derived player and team statistics, such as speed, distance covered, and pass counts, will be assessed mainly through plausibility checks. Good performance here means that values are internally consistent, temporally stable, and broadly compatible with expected football ranges reported in the literature, while still being treated as estimates rather than ground truth [3, 7].

Bird's-eye visualisations and heatmaps will be evaluated qualitatively by examining whether projected player positions and spatial activity patterns are tactically plausible. For example, projected positions should remain within valid pitch boundaries, and heatmaps should broadly reflect expected positional behaviour over a clip. These outputs will also help illustrate how upstream errors affect later stages of analysis.

1.3.2 System-Level Evaluation

At system level, the project will be assessed through the coherence and usefulness of its integrated outputs, including annotated tracking video, bird's-eye views, heatmaps, offside visualisations, and derived statistics. A short end-to-end case study on one representative sequence will be included to demonstrate how the pipeline performs as a complete analysis system rather than as a set of isolated components.

The evaluation will also include explicit analysis of failure cases and error propagation. In particular, the report will examine how errors in detection, tracking, or homography estimation influence downstream modules such as offside analysis, heatmap generation, and statistical summaries. This is important because the value of a modular vision system depends not only on its strengths, but also on understanding where and why failures occur.

Overall, the evaluation strategy combines quantitative model-level evidence, manually constructed labelled subsets for components without existing ground truth, and qualitative assessment of integrated outputs. This provides a realistic and academically defensible basis for judging whether the project has met its objectives, while making its limitations explicit.

1.4 Report Structure

This report is structured so that the reader is guided through the motivation, development, and evaluation of the proposed system, while also providing a clear overview of the project as a whole.

- **Chapter 1 – Introduction.** This chapter introduces the project and its context, outlines the motivation behind the work, states the aims and objectives, and presents the evaluation strategy used to assess the system.
- **Chapter 2 – Background and Theory.** This chapter reviews the relevant literature and technical background needed to understand the project, including the key computer vision and machine learning concepts that inform the chosen methodology.
- **Chapter 3 – Methodology and Implementation.** This chapter describes how the system was designed and implemented. It explains the overall pipeline, the main components used, and the design decisions taken during development.
- **Chapter 4 – Results.** This chapter presents the main outputs produced by the system, including the generated visualisations, event detections, and statistical outputs. It focuses on what the implemented pipeline produces in practice.
- **Chapter 5 – Evaluation.** This chapter evaluates the performance of the individual components and the system as a whole using both quantitative metrics and qualitative analysis. It also discusses failure cases, limitations, and the effect of error propagation across the pipeline.
- **Chapter 6 – Conclusion.** This chapter summarises the main achievements of the project, reflects on its limitations, and outlines possible directions for future work.

Chapter 2

Background and Theory

2.1 Broadcast Football Video Analysis

Automated analysis of football from broadcast video has attracted sustained research interest because television footage is universally available and captures the full spatial and tactical context of a match through a single monocular camera, without requiring dedicated tracking infrastructure or stadium-side sensor installations [22, 14]. In contrast to specialised commercial tracking systems – which supply player-position data from multi-camera or radar installations – a broadcast pipeline must recover all player, ball, and pitch information from a single compressed, moving-camera video stream [6].

This setting introduces several coupled technical challenges. First, the broadcast camera is non-stationary: it undergoes continuous pan, tilt, and zoom in response to play, so the background cannot be treated as a fixed spatial reference and inter-frame motion cannot be attributed to scene objects alone [22]. Second, the ball is extremely small relative to the frame and is frequently blurred, occluded, or temporarily absent, making reliable detection substantially harder than for players [7]. Third, players on the same team share near-identical visual appearance due to uniform colour and similar build, so tracking and team identification must rely on subtle spatial and visual cues [22]. Fourth, image-space measurements are subject to perspective distortion, so quantities such as distance, speed, and tactical positioning are meaningful only after registration to a canonical pitch model [21].

A widely adopted response to this complexity is to decompose the problem into a sequence of specialised components [22, 14, 6]. Object detection and ball detection provide the primary observations, multi-object tracking maintains temporal identity, team assignment infers player affiliation, and pitch registration maps image coordinates into a canonical field representation. Higher-level outputs such as heatmaps, possession estimates, and offside reasoning are then expressed in pitch space rather than image coordinates. The pipeline developed in this project follows this structure, as illustrated in Figure 2.1, and the theoretical basis for each component is developed in the sections that follow.

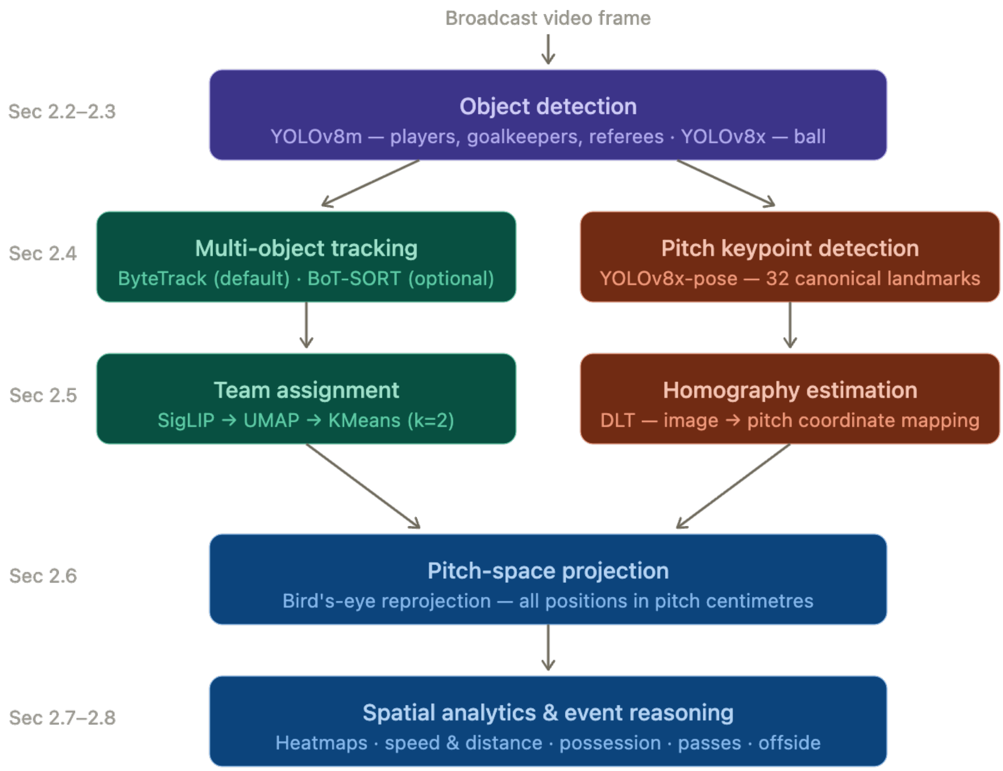


Figure 2.1: Overview of the end-to-end broadcast football video analysis pipeline developed in this project. Teal stages operate in image space; coral stages handle geometric registration; blue stages operate entirely in pitch space. Section references indicate where the theoretical basis for each component is established.

2.2 Object Detection with YOLO

Object detection is the task of simultaneously localising and classifying objects within an image. In broadcast football analysis, the relevant entities are players, goalkeepers, referees, and the ball; for each, a detector must produce a bounding box and a class label. This stage is foundational: missed detections, inaccurate localisation, and class errors propagate directly into tracking, team assignment, homography estimation, and all downstream event reasoning.

Formally, a detector maps an input image I to a set of predictions

$$D = \{(b_i, c_i, s_i)\}_{i=1}^{N}, \tag{2.1}$$

where b_i denotes a bounding box (centre coordinates, width, and height), c_i a class label, and $s_i \in [0,1]$ a confidence score reflecting the detector's certainty that an object of class c_i is present at b_i.

2.2.1 Bounding-Box Localisation and IoU

Localisation quality is measured by the Intersection over Union (IoU) between a predicted box B_p and a ground-truth box B_g:

$$\mathrm{IoU}(B_p, B_g) = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}, \tag{2.2}$$

which takes values in $[0,1]$, with larger values indicating greater spatial overlap (Figure 2.2). A detection is conventionally counted as a true positive only if its IoU with the nearest unmatched ground-truth box exceeds a threshold and the predicted class is correct [13].

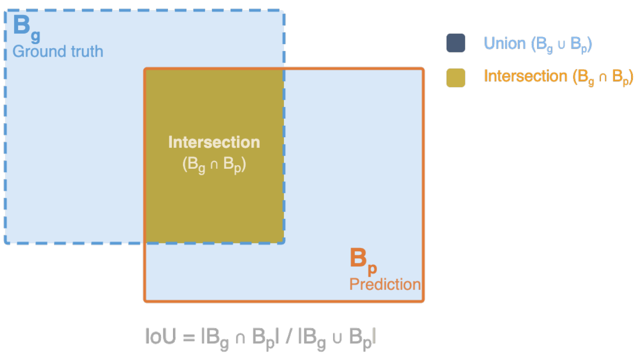


Figure 2.2: Intersection over Union (IoU) between a ground-truth bounding box B_g (blue, dashed) and a predicted bounding box B_p (orange). IoU is the ratio of the intersection area (gold) to the total union area of both boxes, computed as $\mathrm{IoU} = |B_g \cap B_p| / |B_g \cup B_p|$.

Because a detector typically produces multiple overlapping candidates for the same object, Non-Maximum Suppression (NMS) is applied as a post-processing step. NMS retains the highest-confidence box among strongly overlapping predictions and suppresses the remainder [19].

2.2.2 Precision, Recall, and Mean Average Precision

Detection performance is characterised by the precision–recall trade-off. Precision measures the fraction of predicted boxes that are correct; recall measures the fraction of ground-truth objects that are found:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}, \tag{2.3}$$

where TP, FP, and FN denote true positives, false positives, and false negatives respectively. By varying the confidence threshold, a precision–recall curve is obtained; the Average Precision (AP) for a class is the area under this curve, and Mean Average Precision (mAP) is the mean AP across all classes [13].

Two standard operating points are reported in modern benchmarks: mAP_{50}, evaluated at an IoU threshold of 0.50, and $\text{mAP}_{50\text{–}95}$, averaged over thresholds from 0.50 to 0.95 in steps of 0.05. The latter is a stricter criterion because it rewards accurate localisation as well as correct classification. Both metrics are relevant here because the four object classes — players, goalkeepers, referees, and ball — differ substantially in size and detection difficulty, so per-class AP values alongside overall mAP give the most informative picture of detector quality.

2.2.3 The YOLO Architecture

The YOLO (*You Only Look Once*) family formulates detection as a single-stage regression problem: bounding-box coordinates and class confidences are predicted in one forward pass through the network, without a separate region-proposal stage [19]. This makes YOLO well suited to video-based applications where detections must be produced for every frame at high throughput.

Modern YOLO variants, including the YOLOv8 models used in this project, employ an anchor-free detection head [9]. Rather than predicting offsets relative to a set of pre-defined anchor boxes, an anchor-free head directly regresses the box centre, width, and height from each spatial location in the feature map. This simplifies training, removes the need to tune anchor priors for a specific dataset, and improves localisation on small or unusually shaped objects — both relevant to football, where the ball is small and player aspect ratios vary with pose.

Feature extraction is performed at multiple spatial scales through a Feature Pyramid Network (FPN) neck [12], which combines high-resolution, semantically weak features from early layers with low-resolution, semantically strong features from later layers. This multi-scale representation handles the large variation in object size within the same frame, where a close-up referee and a distant background player may coexist.

The confidence score s_i reflects both objectness and class probability. Predictions below a threshold τ are discarded before NMS is applied, introducing a precision–recall trade-off: a high τ reduces false positives at the cost of missed detections, while a low τ recovers more objects at the cost of increased noise. This trade-off is especially consequential for ball detection, where the object is small, frequently occluded, and must be recovered even under low visual confidence; the implications for threshold selection are discussed in Section 2.3.

Figure 2.3 shows a representative detection output from the pipeline, illustrating the variation in object scale across the four target classes.

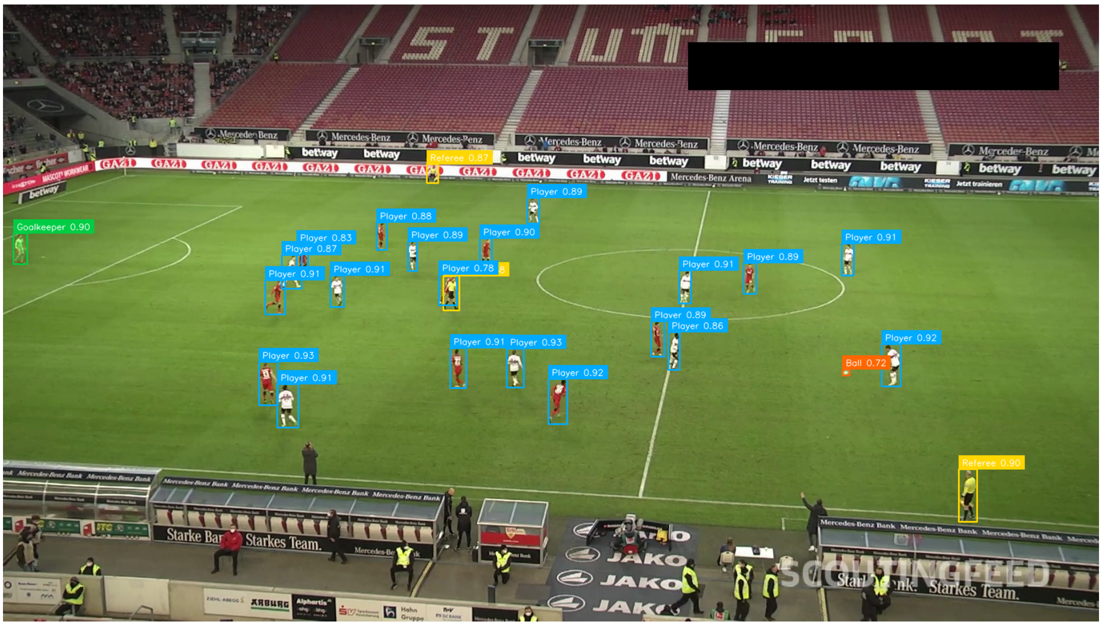


Figure 2.3: Example output of the YOLOv8m object detector on a broadcast football frame. Bounding boxes are shown for all four classes: players (blue), goalkeepers (green), referees (yellow), and ball (red). Confidence scores are shown above each box.

Together, the IoU criterion, NMS, precision–recall evaluation, and the anchor-free YOLO formulation provide the detection-theoretic foundation for Chapter 3 and the evaluation metrics reported in Chapter 5.

2.3 Small-Object Detection and Dedicated Ball Detection

Although the ball is the primary object of interest in football, it is also the hardest to detect reliably in broadcast footage. At typical broadcast resolution, the ball subtends only a few dozen pixels in diameter, moves rapidly enough to produce motion blur, and is frequently occluded by players or absent from the frame entirely [7]. These properties place it firmly in the small-object detection regime, where standard detectors trained jointly on all classes tend to underperform because larger, more visually prominent objects dominate the training signal [22].

The impact of small object size on detection difficulty is quantifiable through the IoU criterion introduced in Section 2.2.1. For a ground-truth box B_g and predicted box B_p, the same absolute pixel displacement causes a disproportionately large drop in $\mathrm{IoU}(B_p, B_g)$ when the object is small, because the intersection area shrinks faster than the union area grows. This makes ball detection acutely sensitive to localisation error, blur, and detector uncertainty in a way that does not apply to larger objects such as players.

A practical response to these challenges is to train a dedicated ball detector with model capacity and inference thresholds tuned specifically for the small-object regime. In particular, operating at a low confidence threshold τ maximises recall at the cost of increased false positives [7]. The resulting candidate set at frame t,

$$C_t = \left\{ \left(\mathbf{b}_t^{(j)}, s_t^{(j)} \right) \right\}_{j=1}^{N_t}, \tag{2.4}$$

where $\mathbf{b}_t^{(j)}$ is a candidate location and $s_t^{(j)}$ its confidence score, will contain the true ball with high probability but also spurious detections from background clutter.

Temporal coherence provides a natural filter for these false positives. Because the ball trajectory is continuous between frames — even during rapid motion, the ball cannot teleport arbitrarily between consecutive observations — a proximity rule can select the most plausible candidate given a previous ball position $\mathbf{b}_{t-1}$:

$$\hat{\mathbf{b}}_t = \arg \min_{\mathbf{b}_t^{(j)} \in C_t} \left\| \mathbf{b}_t^{(j)} - \mathbf{b}_{t-1} \right\|_2, \tag{2.5}$$

subject to a minimum confidence threshold and a maximum permitted inter-frame displacement. The displacement bound acts as a hard plausibility gate: candidates that imply physically implausible ball speeds are discarded regardless of their confidence score. Figure 2.4 illustrates this filtering process on a representative broadcast frame.

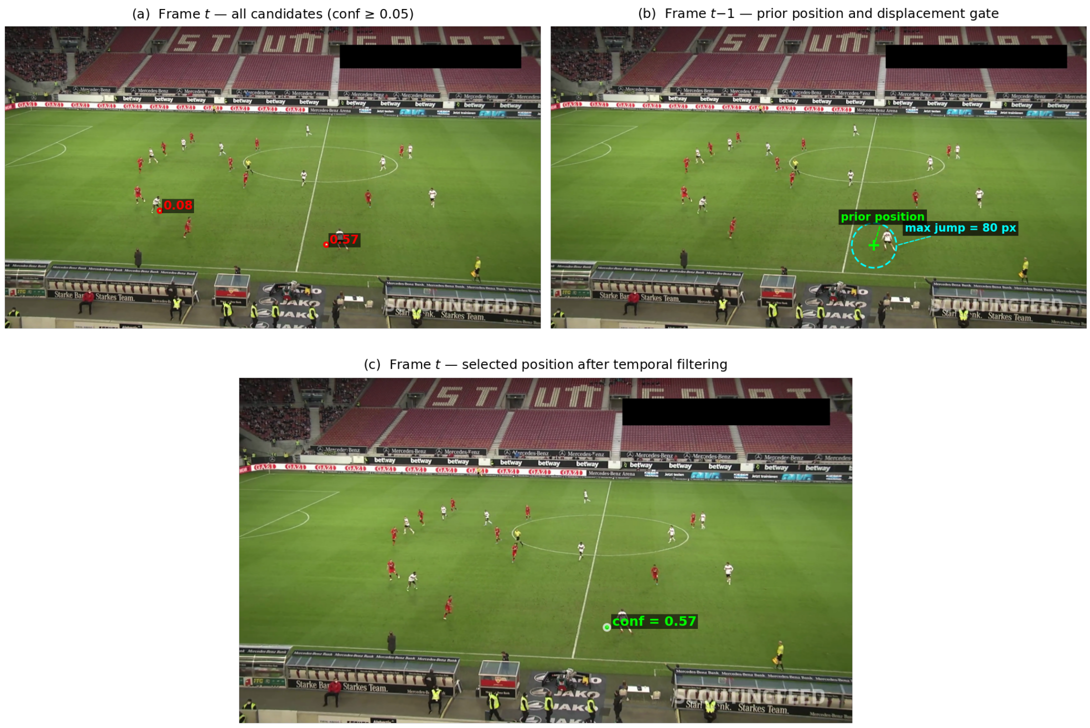


Figure 2.4: Dedicated ball detector temporal filtering on a representative broadcast frame. Top left: two low-threshold candidates ($s \geq 0.05$) detected in frame t, with confidence scores shown in red. Top right: prior ball position $\hat{\mathbf{b}}_{t-1}$ (green cross) and maximum permitted displacement gate of 80 pixels (dashed circle); the lower-confidence candidate at $s = 0.08$ lies outside the gate and is discarded. Bottom: the selected ball position $\hat{\mathbf{b}}_t$ at $s = 0.57$, retained as the closest in-gate candidate, as formalised in Equation (2.5).

This approach differs deliberately from motion-model trackers such as the Kalman filter discussed in Section 2.4.1. A constant-velocity Kalman filter imposes a parametric dynamical model that is poorly suited to ball motion in football, where abrupt accelerations from kicks, rebounds, and deflections are routine. The proximity rule instead exploits only the weakest continuity assumption — that the ball's position changes smoothly between frames — without committing to any particular dynamical model, making it more robust to the sharp trajectory changes that characterise football.

Section 3 describes how these principles are instantiated in the dedicated ball-detection component, and Chapter 5 reports its performance separately from the general object detector.

2.4 Multi-Object Tracking by Detection

Object detection alone produces independent bounding boxes in each frame, but downstream tasks such as team assignment, heatmap accumulation, speed estimation, and offside reasoning all require object identities to remain consistent over time. Multi-object tracking (MOT) addresses this by associating frame-level detections into persistent trajectories [5]. The dominant paradigm in modern MOT is *tracking-by-detection*, in which a detector first produces candidate boxes and a separate association stage links them across frames.

Formally, let

$$D_t = \{d_t^1, d_t^2, \ldots, d_t^{n_t}\} \tag{2.6}$$

denote the set of detections at frame t, and let

$$\mathcal{T}_{t-1} = \{\tau_{t-1}^1, \tau_{t-1}^2, \ldots, \tau_{t-1}^{m_{t-1}}\} \tag{2.7}$$

denote the set of active tracks from the previous frame. The tracking problem is to assign detections in D_t to tracks in $\mathcal{T}_{t-1}$, while creating new tracks for unmatched detections and terminating or buffering tracks that receive no assignment.

2.4.1 Tracking-by-Detection

Tracking-by-detection methods combine motion prediction with data association. Motion prediction estimates where each existing track is likely to appear in the current frame, while data association matches those predictions to the observed detections.

The standard motion model is the linear Kalman filter [10], which maintains a state vector $\mathbf{x}_t$ comprising the bounding-box geometry and its first-order derivatives. The filter operates in two steps. The prediction step propagates the state forward using a constant-velocity model:

$$\hat{\mathbf{x}}_t^- = F\mathbf{x}_{t-1}, \tag{2.8}$$

$$P_t^- = FP_{t-1}F^\top + Q, \tag{2.9}$$

where F is the state-transition matrix, P_t^- is the predicted error covariance, and Q is the process noise covariance. The update step corrects the prediction using the matched detection $\mathbf{z}_t$:

$$K_t = P_t^- H^\top \left(HP_t^- H^\top + R\right)^{-1}, \tag{2.10}$$

$$\mathbf{x}_t = \hat{\mathbf{x}}_t^- + K_t\left(\mathbf{z}_t - H\hat{\mathbf{x}}_t^-\right), \tag{2.11}$$

where H is the observation matrix, R is the measurement noise covariance, and K_t is the Kalman gain, which balances confidence in the model prediction against confidence in the measurement.

Data association is formulated as a bipartite assignment problem. A cost matrix $C \in \mathbb{R}^{m\times n}$ is constructed where C_{ij} measures the incompatibility between track i and detection j, typically as $1 - \mathrm{IoU}$ using the criterion defined in Equation (2.2). The Hungarian algorithm [11] then finds the minimum-cost one-to-one assignment in $O(n^3)$ time, producing matched pairs, unmatched detections that spawn new tracks, and unmatched tracks that are buffered or deleted.

This paradigm scales efficiently to the many simultaneous players in football, but remains sensitive to missed detections, severe occlusion, and the visual similarity between players on the same team.

2.4.2 ByteTrack

ByteTrack improves upon the standard tracking-by-detection pipeline by retaining low-confidence detections rather than discarding them [25]. The motivation is that a detector's confidence score can drop substantially during partial occlusion or motion blur without the underlying object disappearing; discarding such detections prematurely causes unnecessary track fragmentation.

ByteTrack partitions the detector output at frame t into a high-score set

$$D_t^{\text{high}} = \{d \in D_t : s_d \geq \tau_{\text{high}}\} \tag{2.12}$$

and a low-score set $D_t^{\text{low}} = D_t \setminus D_t^{\text{high}}$, where τ_{high} is a confidence threshold. As illustrated in Figure 2.5, association then proceeds in two stages: active tracks are first matched against D_t^{high} using IoU-based cost; tracks that remain unmatched are then compared against D_t^{low}. The intuition is that high-confidence detections establish reliable matches, while low-confidence detections can still recover continuity for temporarily weakened tracks. Tracks that receive no assignment in either stage are buffered for a fixed number of frames before being deleted.

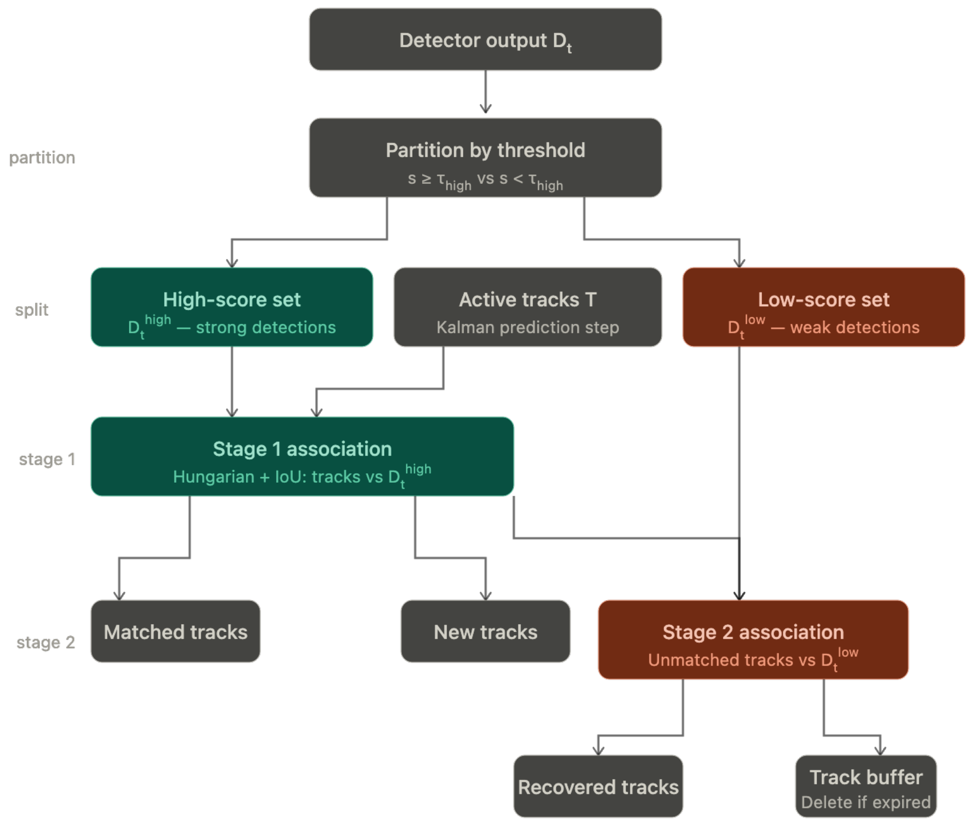


Figure 2.5: ByteTrack two-stage association process. Detector output is partitioned by confidence threshold τ_{high} into a high-score set D_t^{high} and a low-score set D_t^{low}. Stage 1 matches active tracks against D_t^{high} using the Hungarian algorithm with IoU cost; unmatched tracks proceed to Stage 2, where they are compared against D_t^{low}. Tracks that remain unmatched after both stages are held in a buffer and deleted if unrecovered within a fixed number of frames.

In broadcast football footage, where players routinely occlude one another and detector confidence fluctuates during rapid motion, this two-stage strategy reduces identity switches relative to single-stage association. ByteTrack is therefore used as the default tracker in this project.

2.4.3 BoT-SORT

BoT-SORT extends the tracking-by-detection framework with two additional mechanisms: camera-motion compensation and appearance-based re-identification [2]. Camera-motion compensation corrects the Kalman filter's predicted positions for global motion introduced by the broadcast camera, improving association accuracy when the camera pans or zooms rapidly. Appearance-based matching incorporates visual feature similarity as a secondary cost alongside IoU, allowing tracks to be distinguished even when their predicted boxes overlap spatially.

The appearance features used in BoT-SORT are derived from a dedicated re-identification model, in contrast to the general-purpose visual embeddings used for team assignment in Section 2.5. Together, these mechanisms make BoT-SORT more robust than ByteTrack in scenarios with severe camera motion or strong visual ambiguity, at the cost of additional computation. In this project, BoT-SORT is available as a switchable alternative to ByteTrack, with the choice evaluated in Chapter 5.

2.4.4 Tracking Evaluation: HOTA

Tracking quality is assessed using metrics that separately capture localisation accuracy and identity consistency. Multi-Object Tracking Accuracy (MOTA) and Identification F1 (IDF1) are established baselines [5], but both conflate detection and association performance in ways that can obscure the source of errors.

Higher Order Tracking Accuracy (HOTA) was introduced to address this by decomposing overall performance into a detection score and an association score [16]:

$$\mathrm{HOTA}_\alpha = \sqrt{\mathrm{DetA}_\alpha \cdot \mathrm{AssA}_\alpha}, \tag{2.13}$$

where DetA_α measures how well predicted boxes overlap ground-truth boxes at localisation threshold α, and AssA_α measures how consistently predicted identities match ground-truth identities across time. The geometric mean ensures that neither component can dominate: a tracker that localises well but fragments identities, or associates well but misses many objects, will score poorly overall. The final HOTA score is averaged over $\alpha \in [0.05, 0.95]$, making it the primary tracking metric used in this project.

2.5 Unsupervised Team Identification from Visual Features

In broadcast football footage, team identity is not available as an explicit detector output. Players from the two outfield teams must therefore be grouped using visual cues, without labelled examples for the specific match being analysed. This makes team identification a clustering problem rather than a supervised classification task. A useful representation must distinguish jerseys reliably while remaining robust to pose variation, background clutter, scale changes, and illumination differences.

2.5.1 Deep Visual Embeddings

A common approach is to map each player crop to a feature vector in a high-dimensional embedding space. Formally, let I_i denote the image crop of player i. A pretrained visual encoder defines a mapping

$$f : I_i \mapsto \mathbf{z}_i \in \mathbb{R}^d, \tag{2.14}$$

where $\mathbf{z}_i$ captures semantically useful visual information. The aim is that players from the same team produce embeddings that lie closer together than those from opposing teams.

Vision–language models are particularly well suited to this task. SigLIP [24] is trained using a sigmoid-based contrastive loss to align image and text representations over very large datasets, encouraging the encoder to organise its feature space along visually and semantically meaningful axes such as colour, texture, and pattern. Crucially, this alignment is achieved without match-specific supervision: the model has never been explicitly trained on football kits, yet its feature space is sensitive to kit appearance because jersey colour and texture are visually salient properties that correspond to natural language descriptions. This makes SigLIP embeddings more robust than hand-crafted colour features, which cannot capture texture, pattern, shading, or sponsor-logo variation common in professional kits.

The usefulness of an embedding depends strongly on the input region. The jersey torso contains the most discriminative signal for team identification, whereas the lower body, hands, and background contribute noise. Formally, if the full detection box for player i has height h_i, a torso crop I_i^{torso} is defined by retaining only the upper portion of the box. Applying the encoder to I_i^{torso} rather than the full box therefore produces a cleaner embedding for clustering.

2.5.2 Dimensionality Reduction and Clustering

Although informative, embeddings are typically high-dimensional. Clustering directly in a very high-dimensional space can be unstable because Euclidean distances become less discriminative as dimension increases — a phenomenon known as the curse of dimensionality [1]. A standard response is to apply dimensionality reduction before clustering, producing a compact representation that preserves useful neighbourhood structure.

Let $\mathbf{z}_i \in \mathbb{R}^d$ be the embedding of player i and let

$$\phi : \mathbb{R}^d \to \mathbb{R}^{d'}, \quad d' \ll d, \tag{2.15}$$

be a dimensionality-reduction mapping, yielding the reduced representation $\mathbf{y}_i = \phi(\mathbf{z}_i)$.

Uniform Manifold Approximation and Projection (UMAP) [18] is used for this step. UMAP constructs a low-dimensional layout by preserving the local neighbourhood structure of the original space, which contrasts with linear methods such as Principal Component Analysis (PCA) that preserve directions of maximum global variance. Because team identity is reflected more in local appearance similarity between same-kit players than in global variance across all crops, a neighbourhood-preserving reduction is better aligned with the downstream clustering task than a variance-maximising one.

Given the reduced representations $\{\mathbf{y}_i\}_{i=1}^{N}$, k-means clustering [17] partitions them into k groups by minimising the within-cluster sum of squares:

$$\min_{\{\mu_j\}_{j=1}^{k}} \sum_{i=1}^{N} \min_{1 \le j \le k} \|\mathbf{y}_i - \mu_j\|_2^2, \tag{2.16}$$

where μ_j is the centroid of cluster j. Setting $k = 2$ separates the two outfield teams without requiring ground-truth labels. Clustering quality degrades when team kits are visually similar, when crops contain strong background contamination, or when player appearance is significantly affected by motion blur or occlusion — limitations examined in Chapter 5.

Figure 2.6 illustrates the resulting two-cluster structure on a representative match clip, confirming that SigLIP embeddings produce well-separated kit representations in the reduced space.

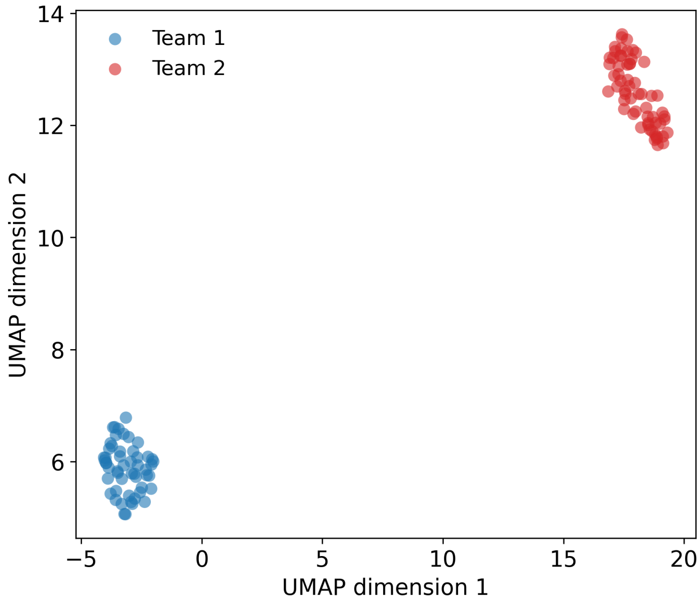


Figure 2.6: UMAP projection of SigLIP torso-crop embeddings for all tracked outfield players in a representative match clip, coloured by k-means assignment ($k = 2$). The clear separation between clusters confirms that the embedding space reliably distinguishes the two teams' kit appearances, supporting the unsupervised identification approach described in Section 2.5.2.

2.5.3 Temporal Label Smoothing

Per-frame clustering predictions are often noisy: a player may be assigned inconsistently across neighbouring frames because of blur, partial occlusion, or crop quality variation. Since team identity is stable over the duration of a track, it is natural to smooth the predicted labels temporally.

Let $\ell_t^{(i)} \in \{1,2\}$ denote the raw predicted team label for track i at frame t. A majority-vote smoother over a sliding window of length W assigns the final label as

$$\hat{\ell}_t^{(i)} = \mathrm{mode}\left(\ell_{t-W+1}^{(i)}, \ldots, \ell_t^{(i)}\right), \tag{2.17}$$

reducing short-term label jitter while preserving the assumption that a player's team is constant within a track. The window length W controls a stability–responsiveness trade-off: larger W suppresses more noise but is slower to correct an erroneously initialised track.

2.5.4 Goalkeeper Team Resolution

Goalkeepers typically wear a kit that differs from both outfield teams, making appearance-based clustering unreliable for this class. A practical alternative exploits spatial context: a goalkeeper is almost always positioned closer to their own team's players than to the opposing team's players.

Let $\mathbf{p}_{\mathrm{GK}} \in \mathbb{R}^2$ be the image-space position of a detected goalkeeper and let $\mathbf{c}_1, \mathbf{c}_2 \in \mathbb{R}^2$ be the image-space centroids of the two outfield-player clusters. The goalkeeper is assigned to team

$$\hat{k} = \arg \min_{j \in \{1,2\}} \|\mathbf{p}_{\mathrm{GK}} - \mathbf{c}_j\|_2, \tag{2.18}$$

substituting a spatial prior for the appearance cue that is unreliable in this case. This assignment is independent of the goalkeeper's jersey colour and requires no additional model.

The team-identification procedure described in this section provides the team labels required by all downstream pitch-space analytics and is evaluated in Chapter 5.

2.6 Pitch Keypoint Detection and Planar Homography

Many outputs of this pipeline — bird's-eye projection, heatmaps, speed estimation, possession analysis, and offside reasoning — require positions to be expressed in a common pitch coordinate system rather than in image pixels. In broadcast football analysis, this is achieved by detecting semantic pitch landmarks and estimating a geometric mapping from the image plane to a canonical pitch model [22, 21].

2.6.1 Pitch Landmark Detection

A football pitch contains a fixed set of semantic landmarks — line intersections, penalty-box corners, and circle tangent points — whose locations are precisely known in a standard top-down pitch representation. Identifying these landmarks in a broadcast frame provides point correspondences between image space and pitch space that can be used for geometric registration.

Formally, let $(\mathbf{x}_i, \mathbf{X}_i)$ denote the i-th correspondence, where $\mathbf{x}_i = (u_i, v_i)^\top$ is the image-space location of landmark i and $\mathbf{X}_i = (x_i, y_i)^\top$ is its known location on the canonical pitch. The role of keypoint detection is to estimate the image-space locations $\{\mathbf{x}_i\}$ reliably enough that they can support homography estimation.

In this project, pitch keypoints are detected using a YOLOv8x-pose model, which extends the object-detection architecture described in Section 2.2 with a dedicated keypoint head. Rather than predicting only bounding boxes and class labels, the pose head additionally regresses the 2D coordinates and visibility confidence of a fixed set of predefined keypoints for each detected instance [9]. Applied to pitch registration, the model is trained to localise 32 canonical pitch landmarks directly from the broadcast frame, producing image-space coordinates together with per-keypoint confidence scores. For homography estimation to be well-conditioned, the retained landmarks must be spatially distributed across the frame rather than clustered in one region, since concentrated correspondences poorly constrain all eight degrees of freedom of the homography. Only landmarks whose confidence exceeds a threshold are therefore retained, discarding poorly localised points that would otherwise degrade the geometric mapping [21].

2.6.2 Homography Estimation

Because the football pitch can be approximated as a plane, the mapping between image coordinates and pitch coordinates is modelled by a planar homography. A homography is a projective transformation represented by a 3×3 matrix H, defined up to a non-zero scalar factor since H and λH define the same transformation for any $\lambda \neq 0$. Using homogeneous coordinates,

$$\tilde{\mathbf{X}} \sim H\tilde{\mathbf{x}}, \tag{2.19}$$

where $\tilde{\mathbf{x}} = (u, v, 1)^\top$ and $\tilde{\mathbf{X}} = (x, y, 1)^\top$ denote image and pitch points in homogeneous form respectively.

Since H has eight degrees of freedom up to scale, at least four non-collinear point correspondences are required to estimate it. With $N > 4$ correspondences the system is overdetermined and is solved in a least-squares sense using the Direct Linear Transform (DLT) [8]. The DLT constructs a linear system $Ah = 0$ from the correspondences, where $h \in \mathbb{R}^9$ is the vectorised form of H. The solution seeks h that minimises $\|Ah\|_2$ subject to $\|h\|_2 = 1$, which is obtained in closed form by taking h as the right singular vector of A corresponding to its smallest singular value.

The validity of this mapping rests on three assumptions: the pitch surface is sufficiently planar; the detected landmarks correspond to the correct semantic points; and the mapping applies only to points on the ground plane, not to objects above it such as players in mid-air. These are standard assumptions in sports-field registration and are appropriate when the goal is spatial reasoning on the playing surface [21].

2.6.3 Reprojection and Bird's-Eye Mapping

Once H has been estimated, any point assumed to lie on the pitch plane can be mapped from image space to pitch space, yielding a bird's-eye representation in which positions can be compared over time without perspective distortion [22]. For player detections, the ground-contact proxy used is the bottom-centre of the bounding box, which best approximates the player's true location on the pitch plane.

Registration quality is quantified by the reprojection error. For a correspondence $(\mathbf{x}_i, \mathbf{X}_i)$, this is

$$e_i = \|\mathbf{X}_i - \pi(H\tilde{\mathbf{x}}_i)\|_2\,, \tag{2.20}$$

where $\pi(\cdot)$ converts homogeneous coordinates to Cartesian coordinates. The mean reprojection error over all N correspondences,

$$\bar{e} = \frac{1}{N}\sum_{i=1}^{N} e_i, \tag{2.21}$$

measures how well the estimated transformation aligns the detected landmarks with their canonical positions. Figure 2.7 illustrates this process on a representative broadcast frame, showing both the detected landmarks used to estimate H and the resulting bird's-eye player projection.

Since homography error propagates to all downstream spatial analysis — shifting projected player positions, distorting heatmaps, and potentially altering offside decisions — accurate field registration is a geometric prerequisite for the entire pipeline rather than an isolated preprocessing step [21]. Pitch keypoint detection and planar homography together establish the geometric foundation for Chapter 3 and the registration evaluation in Chapter 5.

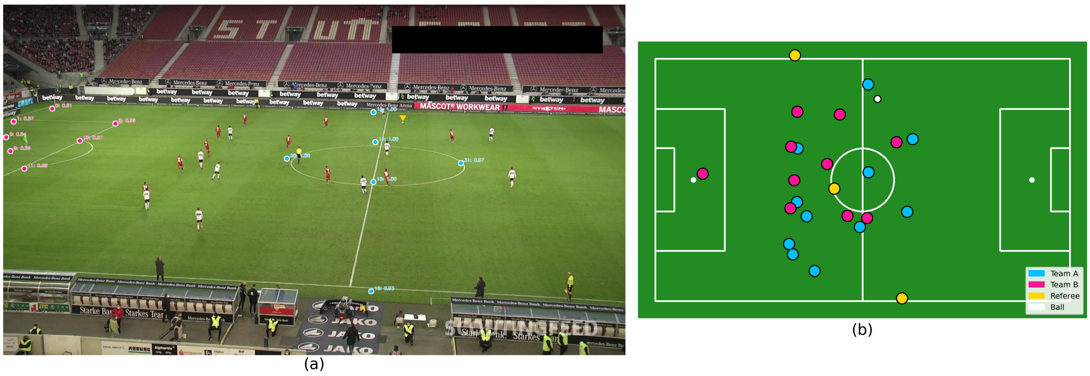


Figure 2.7: Pitch registration on a representative broadcast frame. (a) Detected pitch keypoints overlaid on the broadcast image, labelled with landmark index and confidence score; keypoints are coloured by pitch region — pink for the left half, blue for the centre line, and red for the right half. (b) Corresponding bird's-eye projection after homography estimation, showing all detected players and the ball mapped to the canonical pitch plane and coloured by entity type.

2.7 Spatial Analytics in Pitch Space

Once detections have been registered to a canonical pitch, spatial analysis can be performed in a coordinate system that is independent of camera perspective. Many tactical quantities — player positioning, spatial activity, speed, and distance covered — are meaningful only when expressed on the pitch plane rather than in image pixels [22, 6].

2.7.1 Player Projection onto the Pitch

To analyse player motion on the playing surface, each detection must be represented by a single point on the pitch plane. The most appropriate ground-plane proxy is the bottom-centre of the bounding box, since this best approximates the location at which the player contacts the pitch. Projecting the box centre instead would yield a point that does not correspond to a physically meaningful ground-plane location.

Let $B_t^{(i)}$ denote the bounding box of player i at frame t, and let $\mathbf{q}_t^{(i)} \in \mathbb{R}^2$ denote its bottom-centre anchor in image coordinates. Given the image-to-pitch homography H, the projected pitch-space position is

$$\mathbf{p}_t^{(i)} = \pi\left(H\tilde{\mathbf{q}}_t^{(i)}\right), \tag{2.22}$$

where $\tilde{\mathbf{q}}_t^{(i)}$ is the homogeneous form of the anchor point and $\pi(\cdot)$ converts homogeneous coordinates to Cartesian coordinates. This projection step is the basis of all later spatial analytics: once each player is represented by a pitch-space trajectory $\{\mathbf{p}_t^{(i)}\}_{t=1}^T$, positions can be compared across frames in a common geometric reference and aggregate statistics derived over time.

2.7.2 Heatmap Construction

A player heatmap summarises where that player has been located over a sequence by estimating the spatial density of their projected positions on the pitch. Let $\{\mathbf{p}_t^{(i)}\}_{t=1}^T$ be the projected positions of player i. The pitch is divided into a discrete $M \times N$ grid and an occupancy count is accumulated in each cell:

$$H_i(u,v) = \sum_{t=1}^{T} \mathbf{1}\left[(u,v) = \mathrm{bin}\left(\mathbf{p}_t^{(i)}\right)\right], \tag{2.23}$$

where $\mathbf{1}[\cdot]$ is the indicator function and $\mathrm{bin}(\cdot)$ assigns a pitch position to its grid cell.

Because raw occupancy counts are sparse and visually noisy, the histogram is smoothed by convolution with a Gaussian kernel G_σ:

$$\tilde{H}_i = H_i * G_\sigma, \tag{2.24}$$

where σ controls the spatial scale of smoothing. This is distinct from kernel density estimation (KDE), which evaluates a continuous density function at arbitrary query points; the grid-based accumulation followed by discrete convolution is more accurately described as a Gaussian-smoothed spatial histogram. The result is a continuous-looking density map that summarises spatial tendencies over a clip, making it useful for inspecting player roles, movement concentration, and broad positional behaviour. However, heatmap quality depends directly on the reliability of the projected trajectories from which they are built. Figure 2.8 illustrates this process for a single player, showing the raw trajectory points and the resulting smoothed density map.

2.7.3 Speed and Distance Estimation

Once player positions are available in pitch coordinates, movement statistics are estimated from temporal differences. Let f_s denote the video frame rate in Hz, so that the inter-frame time interval is $\Delta t = 1/f_s$. A finite-difference estimate of velocity for

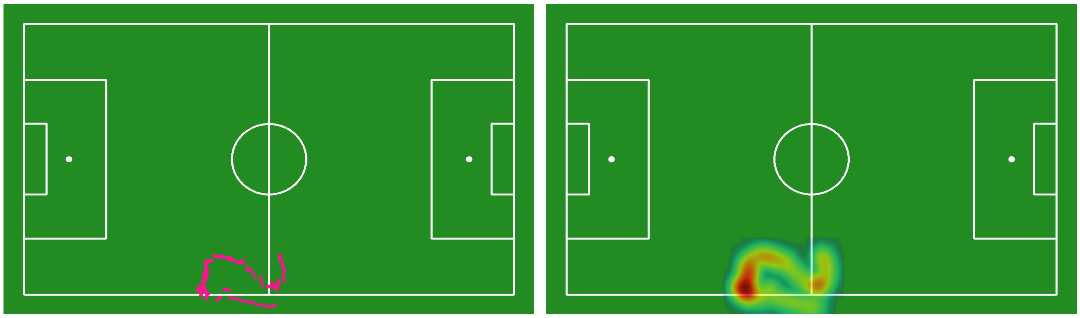


Figure 2.8: Heatmap construction for a single player over a representative clip. (a) Raw projected pitch-space positions, showing the player's trajectory concentrated in the lower-left region of the pitch. (b) Corresponding Gaussian-smoothed heatmap, produced by accumulating occupancy counts on a discrete grid and convolving with a Gaussian kernel as in Equations (2.23) and (2.24).

player i at frame t is

$$\mathbf{v}_t^{(i)} = \frac{\mathbf{p}_t^{(i)} - \mathbf{p}_{t-1}^{(i)}}{\Delta t}, \tag{2.25}$$

with instantaneous speed

$$s_t^{(i)} = \left\| \mathbf{v}_t^{(i)} \right\|_2. \tag{2.26}$$

Total distance covered is approximated by summing successive displacements:

$$d^{(i)} = \sum_{t=2}^{T} \left\| \mathbf{p}_t^{(i)} - \mathbf{p}_{t-1}^{(i)} \right\|_2. \tag{2.27}$$

These estimates are sensitive to noise: small localisation errors in projected positions produce disproportionately large speed values after finite differencing, and identity switches introduce artificial displacement jumps. A standard mitigation is to apply a plausibility cap, discarding speed estimates that exceed a physically reasonable upper bound for human running speed. This acts as a hard gate analogous to the displacement constraint used in ball detection (Section 2.3): observations that imply physically implausible dynamics are suppressed regardless of their origin. Only displacements corresponding to speeds below the cap contribute to the distance accumulation in Equation (2.27). This is consistent with the wider football-performance literature, where running statistics are informative but depend strongly on how the underlying trajectories are obtained and filtered [3, 7].

The spatial analytics described in this section are evaluated in Chapter 5, where their dependence on registration and tracking quality is examined directly.

2.8 Event Reasoning and Match Statistics

Once players and the ball have been projected into pitch space, higher-level match events can be expressed as rules over trajectories rather than as isolated image observations. The main event-level outputs in this project are possession, passes, and offside. These quantities are not observed directly by the detector; instead, they are inferred from tracked spatial relationships over time, making them dependent on both geometric registration and temporal consistency [22].

2.8.1 Ball Possession Estimation

A simple operational definition of possession is based on spatial proximity between the ball and the nearest eligible player. Let $\mathbf{b}_t \in \mathbb{R}^2$ denote the ball position in pitch space at frame t, and let $\mathbf{p}_t^{(i)} \in \mathbb{R}^2$ denote the position of player i, drawn from a candidate set $\mathcal{A}_t$ of on-pitch players excluding referees. The ball holder at frame t is defined as

$$h_t = \arg\min_{i \in \mathcal{A}_t} \left\| \mathbf{b}_t - \mathbf{p}_t^{(i)} \right\|_2, \tag{2.28}$$

subject to the constraint that $\left\| \mathbf{b}_t - \mathbf{p}_t^{(h_t)} \right\|_2 \leq d_{\max}$ for some maximum possession radius $d_{\max}$; if no player satisfies this constraint, possession is unassigned at frame t.

This definition is interpretable and naturally expressed in pitch coordinates. However, it is a heuristic approximation: it can fail when the ball is airborne, when multiple players challenge simultaneously, or when localisation error changes which player appears closest. To reduce short-term noise, possession is confirmed only when the same player remains the nearest valid holder for a minimum number of consecutive frames, preventing rapid switching caused by detector jitter or brief localisation errors.

2.8.2 Pass Detection

A pass is modelled as a stable transition in ball possession from one player to another. Given the possession sequence $\{h_t\}$, a candidate pass event at frame t is detected when

$$h_t \neq h_{t-1}, \tag{2.29}$$

subject to the additional condition that both h_{t-1} and h_t are confirmed holders — that is, each has been assigned possession for at least a minimum number of consecutive frames. This filters out spurious transitions caused by brief detection noise [6].

The moment at which possession changes provides a practical approximation to the instant at which the ball is played, which is central to offside analysis. This makes pass detection not merely a statistic in itself, but a prerequisite for the temporal anchor used in offside reasoning.

2.8.3 Offside Rule Formalisation

Offside is one of the most spatially structured rules in football and a natural target for pitch-space reasoning. Under Law 11, a player is in an offside position if, at the moment the ball is played by a team-mate, they are in the opponents' half and nearer to the opponents' goal line than both the ball and the second-last opponent [20].

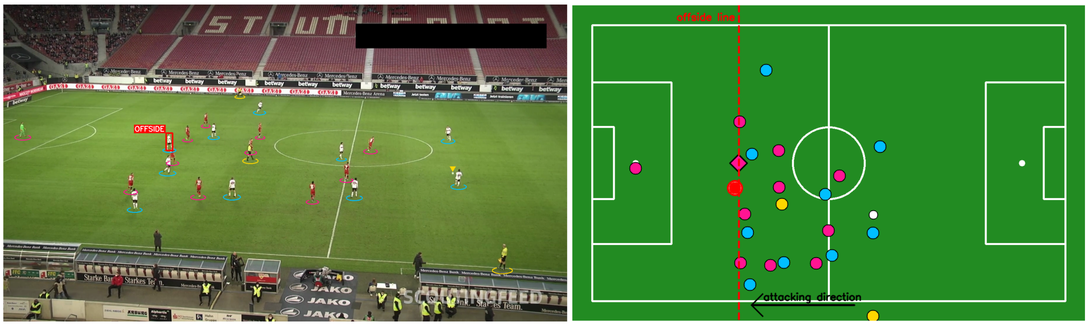
For an automated system, a tractable formulation is to detect *positional offside*: whether an attacking player satisfies the geometric conditions at the relevant moment. Let $x_t^{(a)}$ denote the pitch-space coordinate of an attacking player along the attacking axis, $x_t^{(b)}$ the corresponding ball coordinate, and $x_t^{(o_2)}$ the coordinate of the second-last opponent. The positional offside condition is

$$x_t^{(a)} > \max\left(x_t^{(b)},\, x_t^{(o_2)}\right), \tag{2.30}$$

together with the requirement that the attacker is in the opponents' half. Expressing this as a single max makes the condition more compact and correctly captures that the attacker must be beyond both the ball and the second-last opponent simultaneously. Figure 2.9 illustrates a detected positional offside situation, showing both the broadcast view and the corresponding pitch-space geometry used to evaluate Equation (2.30).

This formulation highlights why pitch-space geometry is essential: offside cannot be judged reliably in image space because perspective distortion alters apparent distances and orderings. By mapping all relevant entities into a common pitch coordinate system, the comparison reduces to a one-dimensional ordering problem along the attacking direction.

A practical requirement is that the attacking direction must be known. In a broadcast pipeline this is not given directly and is instead inferred from the trend of each team's centroid position over time: the team whose centroid moves consistently towards one goal is taken to be the attacking team for that phase of play. This makes



(a) (b)

Figure 2.9: Automated positional offside detection on a representative broadcast frame. (a) Broadcast view with player positions overlaid; one attacking player is flagged as offside (red bounding box). (b) Corresponding bird's-eye projection showing all player positions coloured by team, the offside line (dashed red) computed as $x = \max(x_t^{(b)}, x_t^{(o_2)})$ from Equation (2.30), the second-last defender (diamond marker), and the attacking direction; the flagged player lies beyond the offside line.

offside reasoning dependent not only on accurate player and ball localisation, but also on the stability of team identities and geometric calibration.

The simplified formulation in Equation (2.30) is an approximation. Law 11 depends on the precise moment the ball is played and on whether an offside-positioned player becomes involved in active play — details that a trajectory-based pipeline cannot fully recover. Automated offside analysis should therefore be interpreted as positional offside detection rather than a complete implementation of the law [20]. The accuracy of this approximation is examined in Chapter 5.

2.9 Error Propagation in a Multi-Stage Pipeline

A broadcast-football analysis system is naturally a multi-stage pipeline in which later outputs are produced not directly from the input video but from intermediate estimates generated by earlier modules. Errors therefore do not remain local to the component in which they arise; they propagate forward and affect all downstream analysis. This is an important theoretical consideration in end-to-end sports-video pipelines, where detection, tracking, registration, and event reasoning are tightly coupled [22, 21].

At an abstract level, the pipeline can be viewed as a composition of functions

$$y = f_n(f_{n-1}(\cdots f_2(f_1(x))\cdots)), \tag{2.31}$$

where x is the input video, each f_k represents one processing stage, and y is a final output such as a heatmap, possession estimate, or offside decision. In the present project, the stages $f_1, \ldots, f_n$ correspond to object detection, multi-object tracking, unsupervised team identification, pitch keypoint detection, planar-homography registration, and pitch-space event reasoning. Each stage consumes the structured output of its predecessors rather than the raw video. If the output of stage k is perturbed by some error δ_k, the downstream effect on the final output,

$$\Delta y = f_n(\cdots f_{k+1}(f_k(x_k) + \delta_k) \cdots) - y, \tag{2.32}$$

depends both on the magnitude of δ_k and on the sensitivity of all subsequent stages to it.

In this project, several forms of propagation are especially consequential. A missed detection may break a player trajectory, which can in turn reduce the stability of team assignment and corrupt movement statistics. An identity switch in the tracker may artificially inflate estimated distance covered or create false changes in possession. A registration error in the homography may shift projected player positions, distorting heatmaps and changing the relative ordering of attackers and defenders in offside analysis. An incorrect team label may cause possession, pass counts, or attacking direction to be inferred incorrectly, even if the underlying detections and tracks are geometrically accurate.

This dependency structure means that downstream outputs cannot be interpreted independently of upstream quality. A poor offside decision may reflect not only a weakness in the event logic itself, but also earlier errors in ball localisation, player tracking, team assignment, or pitch registration. Similarly, noisy speed estimates may originate from homography jitter or identity fragmentation rather than from the estimation formula alone.

End-to-end evaluation of a football analysis pipeline should therefore combine component-level metrics with explicit analysis of downstream behaviour [6, 16]. Component-level evaluation identifies where errors originate; system-level evaluation shows how those errors affect the usefulness of the integrated outputs. This motivates the evaluation strategy adopted in Chapter 5, where both individual modules and the final pipeline outputs are assessed.

2.10 Chapter Summary

This chapter has established the theoretical foundation for each component of the broadcast football analysis pipeline. The progression from object detection and dedicated ball detection through multi-object tracking, unsupervised team identification, pitch registration, and pitch-space analytics reflects a deliberate decomposition: each stage reduces a complex vision problem to a well-defined input–output contract, allowing errors to be localised and components to be evaluated independently. The error propagation analysis in Section 2.9 makes explicit that this modularity does not eliminate inter-stage dependencies; it only makes them tractable.

The concepts, metrics, and formalisms introduced here — IoU, HOTA, DLT homography, k-means clustering, temporal smoothing, and positional offside — are used directly in Chapter 3 to describe implementation decisions and in Chapter 5 to assess pipeline performance. Where design choices deviate from the simplest possible approach — such as the use of a dedicated ball detector, torso-only embeddings, or a proximity-based rather than model-based offside trigger — the theoretical justification established here provides the basis for those decisions.

Chapter 3

Methodology and Implementation

3.1 Approach

3.2 Design

3.3 Implementation

Chapter 4

Results

4.1 Experimental Setup

4.2 Findings

Chapter 5

Evaluation

5.1 Analysis

5.2 Discussion

5.3 Limitations

Chapter 6

Conclusion

6.1 Summary

6.2 Future Work

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Appendix A

Additional Materials